

Population Dynamics

Like the piping plover (see Figure 53.1), all populations show some fluctuation in size. Such population fluctuations from year to year or place to place, called **population dynamics**, are influenced by many factors and in turn affect other species. For example, fluctuations in fish populations affect populations of seabirds that eat fish. The study of population dynamics focuses on the complex interactions between biotic and abiotic factors that cause variation in population sizes.

Stability and Fluctuation

Populations of large mammals were once thought to remain relatively stable, but long-term studies have challenged that idea. For instance, the moose population on Isle Royale in Lake Superior has fluctuated substantially since around 1900. At that time, moose from the Ontario mainland (25 km away) colonized the island, perhaps by walking across the lake when it was frozen. Wolves, which rely on moose for most of their food, reached the island around 1950 by walking across the frozen lake. The lake has not frozen over since the early 1950s, and both populations appear to have been isolated from immigration and emigration since then. Despite this isolation, the moose population experienced two major increases and collapses during the last 50 years (Figure 53.18).

What factors cause the size of the moose population to change so dramatically? Harsh weather, particularly cold winters with heavy snowfall, can weaken moose and reduce food availability, decreasing the population size. When moose numbers are low and the weather is mild, food is readily available and the population grows quickly. Conversely, when moose numbers are high, factors such as predation and

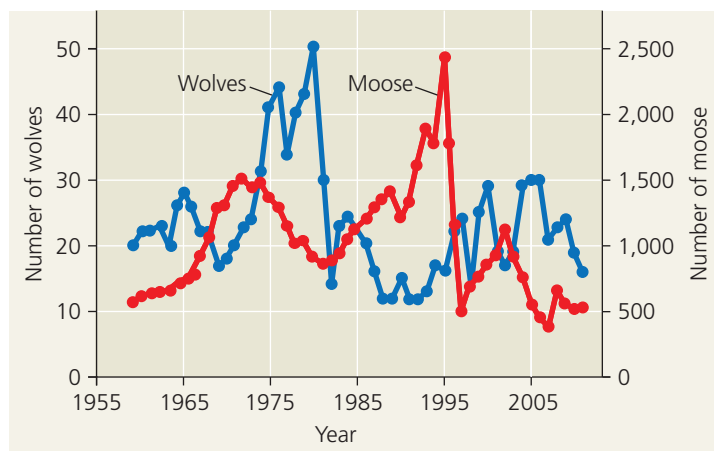
an increase in the density of ticks and other parasites cause the population to shrink. The effects of some of these factors can be seen in Figure 53.18. The first major collapse coincided with a peak in the numbers of wolves from 1975 to 1980. The second major collapse, around 1995, coincided with harsh winter weather, which increased the energy needs of the moose and made it harder for them to find food under the deep snow.

Population Cycles: Scientific Inquiry

While many populations fluctuate at unpredictable intervals, others undergo regular boom-and-bust cycles. Some small herbivorous mammals, such as voles and lemmings, tend to have 3- to 4-year cycles, while some birds, such as ruffed grouse and ptarmigans, have 9- to 11-year cycles.

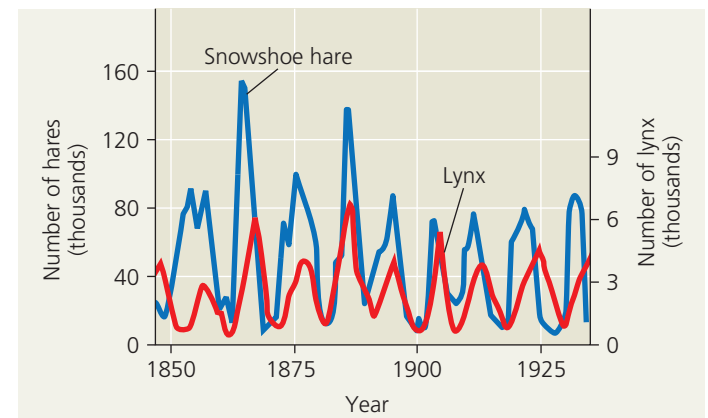
One striking example of population cycles is the roughly 10-year cycling of snowshoe hares (*Lepus americanus*) and lynx (*Lynx canadensis*) in the far northern forests of Canada and Alaska. Lynx are predators that feed predominantly on snowshoe hares, so lynx numbers might be expected to rise and fall with the numbers of hares (Figure 53.19). But why do hare numbers rise and fall in approximately 10-year cycles? Two main hypotheses have been proposed. First, the cycles may be caused by food shortage during winter. Hares eat the

▼ **Figure 53.18** Fluctuations in moose and wolf population sizes on Isle Royale, 1959–2011.



➔ **Mastering Biology BioFlix® Animation: Predator-Prey Interactions**

▼ **Figure 53.19** Population cycles in the snowshoe hare and lynx. Population counts are based on the number of pelts sold by trappers to the Hudson Bay Company.



INTERPRET THE DATA What do you observe about the relative timing of the peaks in lynx numbers and hare numbers? What might explain this observation?